

Geo-spatial Information Science

ISSN: 1009-5020 (Print) 1993-5153 (Online) Journal homepage: www.tandfonline.com/journals/tgsi20

Digital twin-based evaluation of tactical interventions for high-density pedestrian environments

Seungho Yang , Muhammad Usman , Ali Hassan , Gunho Sohn , Petros Faloutsos & Brandon Haworth

To cite this article: Seungho Yang , Muhammad Usman , Ali Hassan , Gunho Sohn , Petros Faloutsos & Brandon Haworth (2026) Digital twin-based evaluation of tactical interventions for high-density pedestrian environments, Geo-spatial Information Science, 29:4, 2868-2886, DOI: 10.1080/10095020.2025.2600195

To link to this article: https://doi.org/10.1080/10095020.2025.2600195

© 2025 Wuhan University. Published by Informa UK Limited, trading as Taylor & Francis Group.

Published online: 23 Dec 2025.

Submit your article to this journal

Article views: 1011

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tgsi20

GEO-SPATIAL INFORMATION SCIENCE
2026, VOL. 29, NO. 4, 2868–2886
https://doi.org/10.1080/10095020.2025.2600195

Taylor & Francis
Taylor & Francis Group

OPEN ACCESS Check for updates

Digital twin-based evaluation of tactical interventions for high-density pedestrian environments

Seungho Yang [a], Muhammad Usman [b], Ali Hassan [c], Gunho Sohn [d], Petros Faloutsos [e] and Brandon Haworth [f]

[a]Department of Urban Engineering, Hanbat National University, Daejeon, South Korea; [b]Ontario Tech University, Oshawa, Canada; [c]Candu Energy Inc, Ajax, Canada; [d]Department of Earth and Space Science and Engineering, York University, Toronto, Canada; [e]Department of Electrical Engineering and Computer Science, York University, Toronto, Canada; [f]Department of Computer Science, University of Victoria, Victoria, Canada

ABSTRACT

Recent urban practice has highlighted the need for safe and efficient pedestrian movement in urban settings. Urban planners and policymakers are tasked with ensuring that individuals can navigate cities while avoiding crowding and minimizing safety risks. Among the various proposed solutions to achieve these planning objectives, enhancing public space adaptability to changing circumstances, such as fluctuations in pedestrian demand and crowding, has garnered significant interest. However, constrained municipal budgets limit the feasibility of large-scale, capital-intensive upgrades to pedestrian infrastructure. As a result, there is an urgent need for a short-term, temporary, and cost-effective strategy to redesign pedestrian public space. This paper introduces a novel tactical urban planning approach, combining evidence-based urbanism. Using a campus digital twin system as an urban simulation platform, this study presents a unique evidence-based planning approach to improve pedestrian spacing on sidewalks by dispersing pedestrians without overly interfering with existing infrastructure. As interventions within existing pedestrian public spaces, one-way sidewalk systems and building entrance control are considered. Through the proposed agent-based simulation of the campus digital twin system, the optimization of their application and the demonstration of its effectiveness are achieved. The outcome of the simulations verifies that these measures can enhance pedestrian spacing by dispersing them without causing substantial adverse effects. While these findings stem from experiments conducted on a particular campus, limiting their generalizability, they still hold significance as they verify the potential for practical implementation of tactical urban planning concepts utilizing a digital twin system for urban environments.

ARTICLE HISTORY
Received 25 December 2023
Accepted 3 December 2025

KEYWORDS
Tactical interventions; evidence-based planning; agent-based model; pedestrian spacing; crowd management; digital twin

1. Introduction

Providing citizens with a high-quality walking environment is essential in contemporary cities (Gehl 2013). In public spaces that attract large crowds, effective management of high-density pedestrian flows is critical for safety, comfort, and the overall performance of public space (Dufour et al. 2025; Schneider et al. 2021; Zhang et al. 2023). Although several studies have explored different urban planning interventions for crowding and pedestrian management, there is a noticeable lack of attempts to combine tactical urbanism with evidence-based planning approaches. In this paper, we aim to address this gap by proposing a tactical urban planning approach to improve pedestrian movement in high-density urban settings, utilizing evidence-based planning principles and implementing a simulation in a digital twin of a university campus as a case study.

Urban planning and design, which involves technical and political processes aimed at shaping the physical layout, land use, and built environment of human settlements (Taylor 1998), has evolved over time, giving rise to various innovative approaches. One such trend is tactical urban planning, which is an agile, low-cost, and scalable method that facilitates rapid implementation and iterative adjustments to urban design. This tactical approach addresses the limitations and challenges of traditional urban planning that may be hindered by institutional capacity and complex procedures (Alonso et al. 2018). Evidence-based planning, on the other hand, is an essential aspect of urban planning that ensures interventions are grounded in robust data and analysis for evaluating alternative and optimizing planning scenarios, forecasting the effects of diverse interventions on urban spaces (Alonso et al. 2018; Cheshmehzangi 2016; Chung and Choo 2011; Davis et al. 2006; Moonen et al. 2012; Mueller, Klein, and Hof 2018). Recently, digital twin technology, a powerful tool

CONTACT Seungho Yang sh.yang@hanbat.ac.kr

© 2025 Wuhan University. Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

within evidence-based planning, has emerged as a means to monitor and simulate cities under various conditions (Batty 2018).

In urban planning and design, sidewalks are the primary infrastructure enabling pedestrian mobility. In commercial districts or public parks, walking on sidewalks can itself be recreational; however, in most settings sidewalks primarily serve a utilitarian mobility function (Kang et al. 2017; Southworth 2005; Yin et al. 2023). From this functional perspective, the effective management of high-density pedestrian flows on sidewalks is essential. Such management ensures comfortable walking conditions in everyday circumstances and can prevent crowd-related safety accidents during temporary events (Asadi-Shekari, Moeinaddini, and Shah 2014; Still et al. 2020). Moreover, as widely demonstrated during the COVID-19 pandemic, when many cities emphasized physical distancing, high-density pedestrian management also plays a critical role in reducing transmission risk during infectious-disease outbreaks (Chu et al. 2020; Li, Yin, and Chen 2023; Psyllidis et al. 2023; Xiao et al. 2022).

In this study, among various approaches to high-density pedestrian management, we focus on pedestrian spacing. As seen in critical cases such as the strict lockdown policies during the COVID-19 period, reducing the number of pedestrians through demand control can be the most effective measure in emergency situations. However, this is a very exceptional case. In this study, we discuss ways to control pedestrian density while accepting the demand for sidewalk use as it is in everyday life. We define this practice as pedestrian spacing, which refers to reducing close contact among pedestrians to lower the pedestrian density experienced by individuals, leading to comfortable and safe walking.

Despite numerous studies examining urban planning interventions for crowding and pedestrian management, research that combines tactical urbanism with evidence-based planning remains limited. This paper proposes a tactical urban planning method for improving pedestrian spacing by dispersing pedestrians without major disruptions to existing infrastructure. We employ evidence-based planning principles and implement a novel simulation at digital twin of a university campus as a case study to assess the effectiveness of these measures.

Considering this context, our study introduces a predictive approach that incorporates evidence-based planning and tactical urbanism principles using simulations. We demonstrate this method by identifying temporary interventions in urban spaces to reduce close encounters and alleviate crowding. By applying candidate interventions in a virtual urban environment, we evaluate their impact through simulation. This enables us to assess the efficacy of these interventions in minimizing close encounters among pedestrians on sidewalks and underscores the value of digital-twin – based simulation for urban planning.

This paper is organized as follows: Section 2 provides a review of relevant literature, discussing the principles of tactical urban planning, evidence-based planning, interventions for pedestrian spacing, and digital twin technology. Section 3 outlines the methodology employed, detailing the steps taken to develop and implement the digital twin creation and pedestrian simulation. Section 4 presents experimental designs for simulating the intervention of pedestrian movements. Section 5 discusses the results and analysis of the simulation, highlighting the advantages and limitations of the proposed urban planning approach. Finally, Section 6 concludes the paper, summarizing the key findings and offering suggestions for future research and implementation.

2. Background

2.1. Urban planning and design with tactical urbanism for pedestrian management

As many cities transition from car-centric to pedestrian-oriented planning, they are working to secure more space for walking. A principal approach is to enlarge the existing pedestrian realm – widening sidewalks or implementing road diets that reallocate motor-vehicle lanes to pedestrian space. While such physical retrofits are demonstrably effective at improving pedestrian spacing (Asadi-Shekari, Moeinaddini, and Shah 2014; Noland et al. 2015), they are constrained by limited right-of-way and substantial capital costs. Moreover, they are ill-suited to rapid, temporary responses for short-duration events or future pandemics such as COVID-19. For these reasons, alongside long-term upgrades to pedestrian infrastructure, cities should also consider operational strategies that improve the performance of existing facilities – an area in which the principles of tactical urbanism can be productively applied.

Tactical urbanism, an urban planning approach that prioritizes adaptability amid rapid social changes, is another crucial trend. In contrast to traditional urban planning that emphasizes long-term changes (Law, Azzali, and Conejos 2021), tactical urbanism focuses on short-term, temporary, small-scale, low-budget projects with active community involvement (Graziano 2021; Pradifta et al. 2021). Due to its grassroots origins, tactical urbanism has been referred to by various names, including guerrilla urbanism, pop-up urbanism, Do It Yourself (DIY) urbanism, and temporary urbanism (Andres 2013; Ferreri 2015; Graziano 2021; Lydon and Garcia 2015). It is often used to assess the effects of urban space modifications through temporary transformations employing strategic painting, paving, and even potted plants. Few studies have

explored the application of tactical urbanism to crowding and pedestrian management, typically focusing on specific cases or treating it as a precursor to long-term changes (Barata and Fontes 2017; Gregg et al. 2022).

This study suggests more subjective tactical urban planning components and highlights the need for a stronger foundation for such planning actions. Although tactical urbanism inherently serves as a trial for more permanent urban space changes, evidence-based planning using simulations or technological methods can make the approach more scientific and objective. Big data, urban digital twins, simulations, and other advanced technologies can enhance the capacity to adapt to unforeseen circumstances (Angelidou et al. 2017).

2.2. Countermeasures for pedestrian spacing in high-density pedestrian environment

From this point forward, we discuss the countermeasures to enhance pedestrian spacing in high-density pedestrian environments without constraining pedestrian demand. In doing so, we draw on a range of approaches employed during the COVID-19 period. At that time, various measures were implemented or considered to minimize close contact and promote physical distancing. From this broader set, we identify those expected to be applicable in high-density pedestrian settings with minimal public resistance. This subsection will discuss two such strategies: one-way sidewalks and building entrance control.

The concept of one-way sidewalks was introduced to reduce close encounters between pedestrians on narrow pathways, with several cities and large retailers either implementing or contemplating this strategy (Cardiff News Room 2020; City of Westmount 2020; Oxford City Council 2020; Pearson 2020; Romero, Stone, and Ford 2020; Storeys 2020). It is important to recognize, however, that close contact between pedestrians may still transpire even under one-way conditions, casting doubt on the measure's effectiveness. Nevertheless, if less congested alternative routes are accessible near crowded sidewalks, one-way streets can facilitate pedestrian dispersion and enhance physical distancing. This principle has been explored in traffic engineering, where one-way traffic systems have been shown to alleviate congestion by dispersing vehicles (Karimi et al. 2022; Zhang et al. 2020). We propose that this principle could also apply to pedestrian traffic.

Amidst the COVID-19 pandemic, many buildings have adopted entrance screening to regulate the number of entrants and measure their body temperatures (Ruki and Wulandari 2021). In certain instances, only a limited number of entrances are operational, while the others remain closed to streamline management. Although this technique primarily aims to prevent disease transmission within buildings, it can also be employed to promote physical distancing on sidewalks. By shutting down entrances near busy sidewalks and only opening those adjacent to less frequented walkways, this approach can effectively distribute pedestrians. In a similar vein, previous research has demonstrated that theme parks can mitigate visitor congestion by modifying the placement of attractions (Ohtani, Sawada, and Nogami 2010). We suggest that controlling building entrances can similarly disperse pedestrians by adjusting the location of available entry points.

These countermeasures, along with others tailored to specific contexts, demonstrate the potential for innovative strategies to minimize close contact between individuals and promote pedestrian spacing in high-density pedestrian environments. However, the effectiveness of these measures may vary depending on the specific circumstances, and further research is needed to determine the best practices for maintaining comfort and safe environments in various settings. In this study, we have employed both one-way sidewalks and building entrance control countermeasures to assess their impact on minimizing close contact and enhancing pedestrian spacing in urban environments.

2.3. Evidence-based planning, simulation, and digital twin

In urban planning, predicting the future is crucial. Simulation is a key approach used to forecast future scenarios and is essential for evidence-based planning. To achieve this, various simulation models or tools have been developed for different urban aspects such as land use, transportation, demographics, microclimate, disaster, and landscape (Champion 2001; Davis et al. 2006; Mueller, Klein, and Hof 2018; Tyrna and Hochschild 2010; Urech, Mughal, and Bartesaghi-Koc 2022; Walliss 2018).

A digital twin is a virtual world that is based on a 3D model connected to the real world. It is also considered a simulation process because real data is applied to the virtual world to provide feedback to the real world (Deren, Wenbo, and Zhenfeng 2021). The urban digital twin, a concept that applies digital twin technology to the urban space (Yang and Kim 2021), can replicate and predict changes in the urban space and the behaviors of people in the urban environment. The urban digital twin has a differentiation from other similar simulation platforms in that it provides simulations using real-time Internet of Things (IoT) sensor data, Closed-Circuit Television (CCTV) data, and other various big data, and actively utilizes 3D visualization as needed for this purpose (Jeddoub et al. 2023; Lehtola et al. 2022; Nochta et al. 2021). Ultimately, as an urban-related simulation platform,

urban digital twin technology is a useful tool for evidence-based planning.

In fact, the urban digital twin is being utilized in many cities around the world for the operation and management of cities. In these actual cases of urban digital twins, including cities such as Singapore, Tokyo, London, and Seoul, the urban digital twin demonstrates differences in its implementation level and functionality in the aspect that it includes numerous diverse components related to the urban space. The level of detail in replicating the physical environment also varies, and its functions range widely from simple visualization to monitoring and prediction (Jeong et al. 2023).

3. Methodology

Figure 1 shows an overall workflow illustrating the proposed digital twin-based simulation framework for tactical urbanism. This approach is designed to evaluate the impact of simulated sidewalk modification on pedestrian spacing and pedestrian flow efficiency. A small-scale area of interest ("Real World" block with yellow color shown in this figure) is identified for improving pedestrian movement. The workflow with green color showcases the steps of proposed simulation in digital twin involved in setting goals, brainstorming countermeasures, conducting simulations, and optimizing interventions based on feedback. The outcomes of simulation are taken in "Physical Intervention" block with yellow color for realizing the simulation-based decision to modify pedestrian movements. The framework can be applied to various urban settings, with this specific example focusing on a university campus under high-density pedestrian conditions.

3.1. Possible interventions for pedestrian spacing on sidewalk

NACTO (2020) introduced a variety of strategies aimed at maintaining physical distancing in public spaces, such as sidewalks. These strategies differ in their underlying principles and intensity levels. Essentially, the degree of physical distancing depends on the number of people occupying a given space. For instance, when pedestrian traffic on sidewalks becomes denser, maintaining adequate distance between pedestrians becomes increasingly difficult. As a result, temporary measures for promoting pedestrian spacing generally focus on reducing the density of users in a particular space.

In this study, we explored methods to enhance pedestrian spacing without physically modifying existing urban infrastructure or reducing the number of people using urban spaces. To decrease localized density while maintaining the same overall space size and user count, it is necessary to distribute pedestrians more evenly throughout the area. Upon reviewing various interventions, we discovered that specific strategies, such as one-way sidewalks and building entrance controls, can effectively disperse pedestrians and improve pedestrian spacing. These interventions impose minimal restrictions on pedestrians, requiring only slight inconveniences, such as taking a short detour. Additionally, they can be implemented temporarily, tactically, and at a low cost without necessitating significant construction or changes to the built environment.

First, one-way sidewalks were proposed as a means to minimize close encounters between pedestrians on narrow walkways, and several cities and large retailers have either implemented or considered this approach (Cardiff News Room 2020; City of Westmount 2020; Karedes 2020; Oxford City Council 2020; Pearson

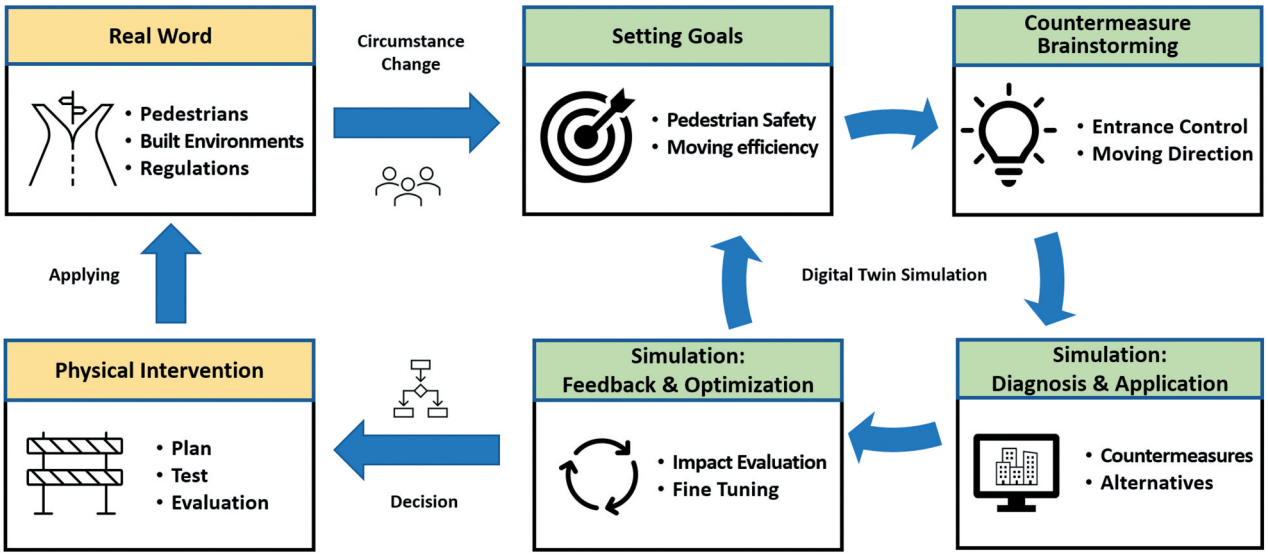


Figure 1. A conceptual workflow illustrating the general stages of digital twin-based simulation framework for tactical urbanism.

2020; Romero, Stone, and Ford 2020; Storeys 2020). However, it is worth noting that close contact between pedestrians can still occur even in one-way situations, so the effectiveness of this measure remains unclear. Yet, if less-trafficked alternative routes are available near heavily congested sidewalks, one-way streets can help to disperse pedestrians and improve pedestrian spacing.

Second, during the COVID-19 pandemic, many buildings have implemented entrance screening to monitor the number of entrants and their body temperatures (Ruki and Wulandari 2021). In some cases, only a few selected entrances are open, while the rest remain closed to streamline management. Although this method is primarily employed to curb disease transmission within buildings, it can also be utilized to enhance pedestrian spacing on sidewalks. By closing entrances adjacent to busy sidewalks and only opening those next to under-utilized walkways, this strategy can effectively disperse pedestrians.

3.2. Pedestrian spacing simulation using digital twin campus

To demonstrate the effects of the interventions described earlier and suggest their suitable applications, this study utilized agent-based modeling. Pedestrian simulation was conducted on the digital twin of York University Keele Campus, located in Toronto, Canada. This campus spans over 450 acres, has over 90 buildings, and hosts over 60,000 members, including faculty, staff, and students. With two subway stations located inside the campus and many bus lines passing through it, the campus has excellent public transit conditions, leading to high pedestrian activity. In this context, we have chosen the York University Keele Campus for our study because it allows us to easily apply our a priori knowledge of the campus layout and pedestrian transportation dynamics. Additionally, the selected campus features a large and dynamic urban environment, which makes it an ideal setting to assess the impact of our pedestrian movement intervention model. By studying a complex environment like this, we can demonstrate the applicability and effectiveness of our approach in real-world situations. Thus, York University Keele Campus is a suitable place for pedestrian crowd simulation. The pedestrian movements of university members were simulated on the 3D campus digital twin according to the experimental design.

For the simulations, the study extracted nine main buildings located in the center of the campus and the sidewalks connected to them, as shown in Figure 2. These include a subway station, a student center, classroom buildings, and department buildings. The need for physical distancing interventions is minimal for sidewalks located on the outskirts of the campus (outside the focus area of this paper) because of relatively low pedestrian volume. Therefore, the nine selected buildings were deemed sufficient to demonstrate the effect of interventions by simulation.

3.3. Pedestrian simulation

Given the importance of pedestrian spacing in high-density pedestrian environment, numerous studies have investigated the issue, with most proposing computer vision-based detection technology to monitor physical distance among people (Madane and Chitre 2021; Pari, Vasu, and Geetha 2020; Shao et al. 2021; Su et al. 2021; Yang et al. 2021). These studies used video data to detect individuals who did not adhere to the ordinary distance standards (e.g. 1 m or 2 m). Furthermore, some studies employed agent-based models to analyze pedestrian spacing. Harweg, Bachmann, and Weichert (2021) utilized an agent-based model to measure close contact exposure time between agents and suggest a suitable user density for safe pedestrian environment, while Romero, Stone, and Ford (2020) used exposure time to predict the efficacy of a one-way sidewalk intervention in an indoor space via mathematical modeling. As our study also simulated pedestrian movement to demonstrate the effectiveness of proposed interventions in enhancing pedestrian spacing, the duration of close contact between agents (within 2 m) was calculated for each agent. While 2 m served as the baseline proximity threshold, the simulation allows users to specify alternative thresholds to reflect different risk levels and design standards.

In this section, we provide a detailed overview of the simulation platform used in our study. The simulation cycle consists of several main components, as depicted in Figure 3. To develop the pedestrian simulator, we utilized the Unity3D Gaming Engine (Unity, 2021). By integrating the Building Information Modeling (BIM) models of the university campus, we were able to recreate a realistic digital environment, as shown in Figure 1. The platform includes an interactive User Interface (UI) that allows users to configure Scenarios and run simulations (Figure 4). A scenario configuration includes several control elements that need to be set before running a simulation, such as (a) setting constraints for the walkways, (b) enabling/disabling building entrances, (c) setting the overall simulation run time, and (d) selecting the origin and destination buildings for pedestrians. The UI also allows for setting pedestrian-related parameters such as (e) pedestrian spawning interval types, (f) spawning rate, (g) social distancing constraints, and (h) alternate walking routes. As input data for the simulation, users can add as many scenario configurations as desired and run the simulation once all scenarios

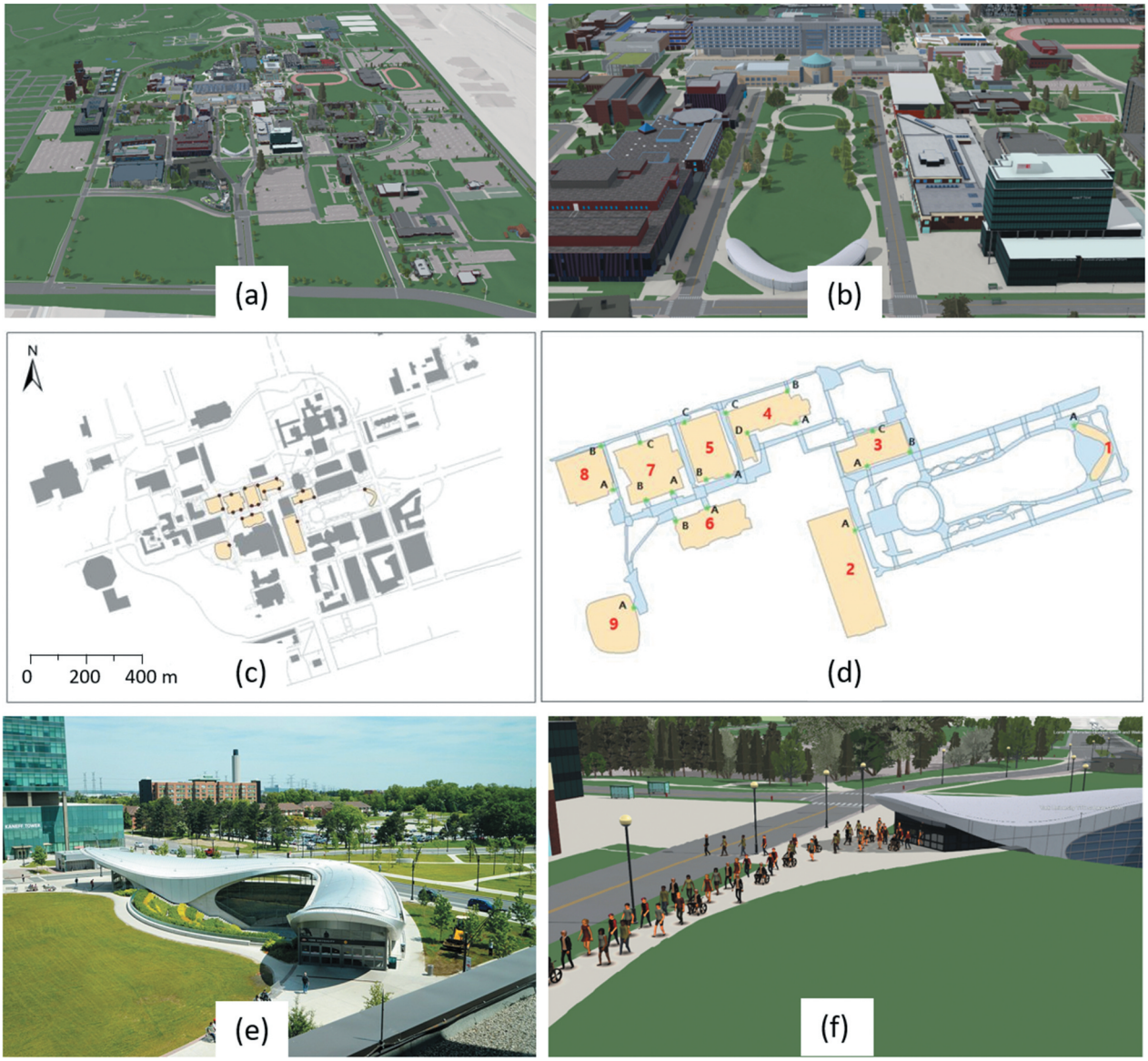


Figure 2. Visual comparison of York University's Keele campus and its digital twin, with a focus on the simulation experiment area: (a) Comprehensive view of the Keele digital twin campus; (b) Highlighted section of the Keele digital twin campus tested in this paper; (c) 2D map of the Keele campus displaying 9 buildings, differentiated by a yellow hue; (d) Enlarged area for the simulation, showcasing buildings in yellow (numbered in red) and walkway paths in blue; (e) A photo of the York University subway station used in the simulation; and (f) A captured scene of simulated agents navigating around the York University subway station.

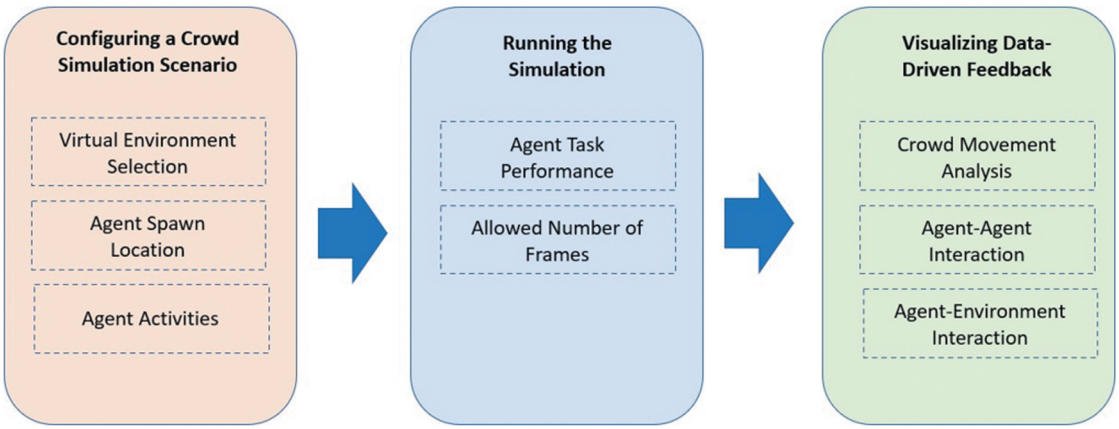


Figure 3. Key elements of a crowd simulation cycle.

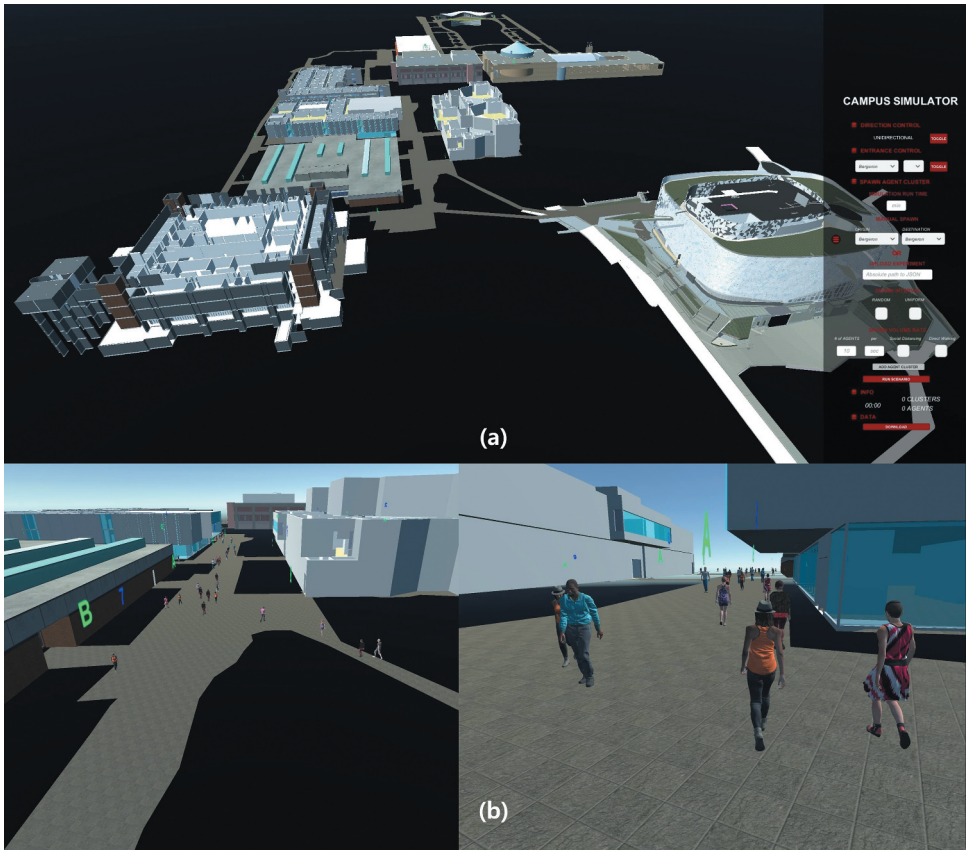


Figure 4. The pedestrian simulation tool to run campus-wide simulations: (a) The user interface of the simulation tool, and (b) Sample pedestrian simulation snapshots.

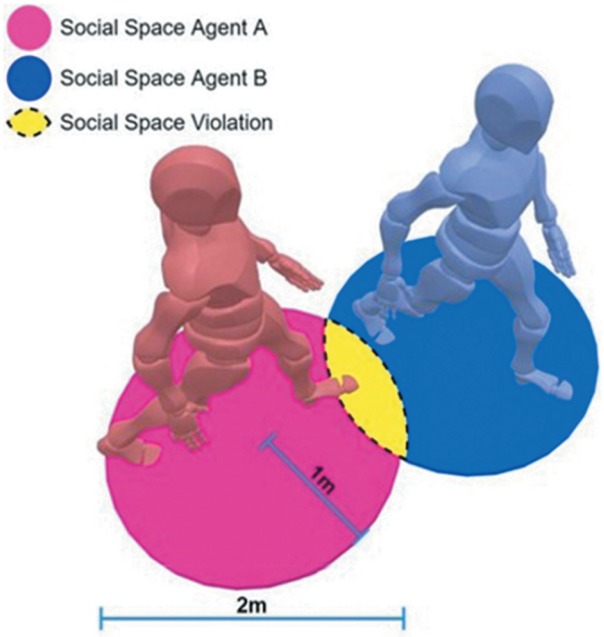


Figure 5. An illustration of an agent's social space and its violation.

are added. During the simulation or upon completion, users can download the logged data of pedestrian trajectories in a CSV format as output data from the simulation. Figure 5 shows an example of social distancing constraints.

Pedestrian simulations aim to represent human-building and human–human interactions in virtual environments over time. To achieve this, researchers have explored various pedestrian navigation and collision avoidance techniques, each mimicking different aspects of human-like movements (Kapadia et al. 2015; Pelechano et al. 2016). This study uses the Predictive Avoidance Model (PAM) (Karamouzas et al. 2009), which is a physical force-based approach that uses attraction and repulsion forces to model individual pedestrian movement and interactions. PAM applies forces that push pedestrians toward their destination and away from obstacles or other pedestrians. It is important to emphasize that PAM operates at a microscopic level, meaning that it simulates the behavior of individual pedestrians rather than treating the crowd as a homogeneous entity. In PAM, each pedestrian autonomously makes decisions based on its immediate surroundings and goals. At its core, PAM considers a range of factors for each individual pedestrian when determining its next position in space. These factors include the pedestrian's current position, the location of its target goal, and the presence of obstacles or other pedestrians in its vicinity. By continuously assessing its surroundings, each pedestrian dynamically adjusts its movement to navigate toward its goal while avoiding collisions with obstacles and other agents.

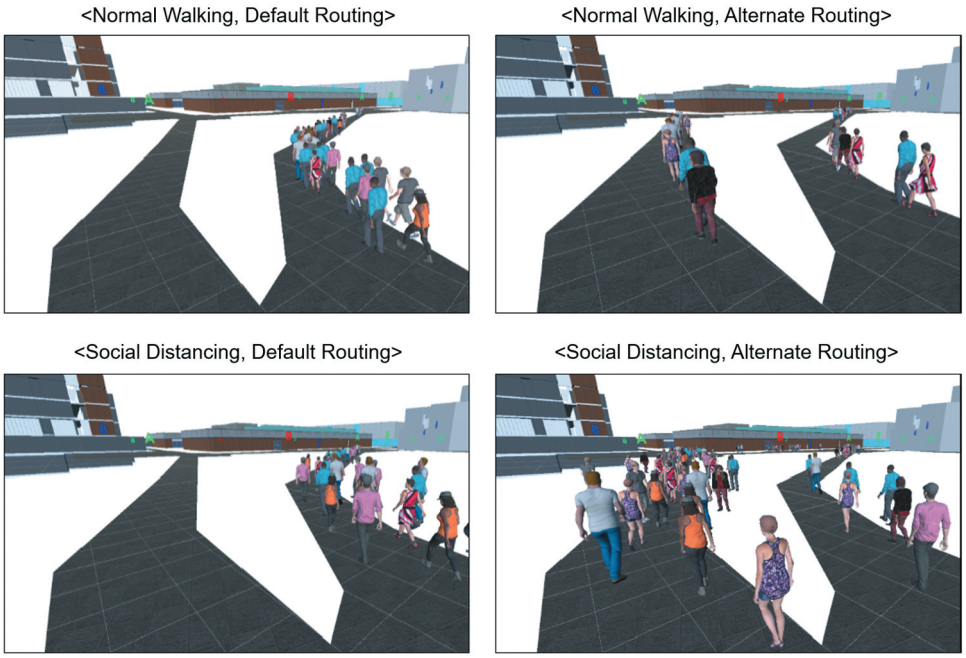


Figure 6. A snapshot of a simulation where pedestrians are moving from building 9 to 7 (as per Figure 2) in a unidirectional walkway setting.

This microscopic approach allows us to capture the complexity of pedestrian behavior in crowded environments. Rather than relying on predefined rules or trajectories for the entire crowd, PAM enables each pedestrian to make independent decisions based on real-time information, resulting in emergent crowd dynamics that closely resemble real-world scenarios. By incorporating the microscopic nature of PAM into our simulations, we ensure that every individual pedestrian behaves autonomously, responding to its environment in a realistic manner. In a related example, PAM has been utilized in a study that simulated potential collisions between pedestrians and sidewalk delivery robots during their navigation (Hassan et al. 2023). Similar force-based techniques have been used previously in architectural applications to evaluate human-building interactions (Schaumann et al. 2020; Usman et al. 2019a, 2019b, 2020, 2021).

To introduce unidirectional and bi-directional walking in the simulations, constrained Navigation Mesh (NavMesh) links are used. NavMesh links create navigable connections between two walkway segments on a NavMesh and can include a permissible walking direction constraint (e.g. Left-to-Right or Right-to-Left). By default, the walkway segments are set to bi-directional walking. To enable alternate path routing, navigation area costs are used to define the relative difficulty of walking across a specific walkway segment. A lower cost indicates a preferred path during heuristic pathfinding. Initially, pedestrians are assigned the shortest path to their destinations. However, as the shortest path becomes more crowded, alternate routes become cheaper and thus more appealing to pedestrians, even if they are longer than the shortest path.

While Unity is primarily known for its capabilities in gaming controls, logic, and 3D graphical rendering, it offers flexibility through its scripting capabilities. Leveraging Unity's scripting languages (e.g. C#), we integrated the PAM simulations seamlessly into the Unity environment. This integration enables us to harness the visualization and interaction capabilities of Unity while leveraging the realistic pedestrian movements and interactions provided by PAM. To summarize, while Unity serves as the platform for visualization and user interaction, the core pedestrian simulation is driven by the PAM, with additional functionality facilitated through constrained navigation meshes within Unity. This approach allows us to create realistic pedestrian simulations that consider real-world origin-destination and pedestrian flow rates.

We conducted a series of experiments using the pedestrian simulation platform discussed earlier to explore the effect of interventions on improving pedestrian spacing between pedestrians in public spaces. A sample snapshot of the pedestrian simulation is depicted in Figure 6.

4. Experimental design

4.1. System overview

Figure 7 depicts a process for creating various scenarios of pedestrian movement and assessing their effects on simulated designs of pedestrian motion interventions.

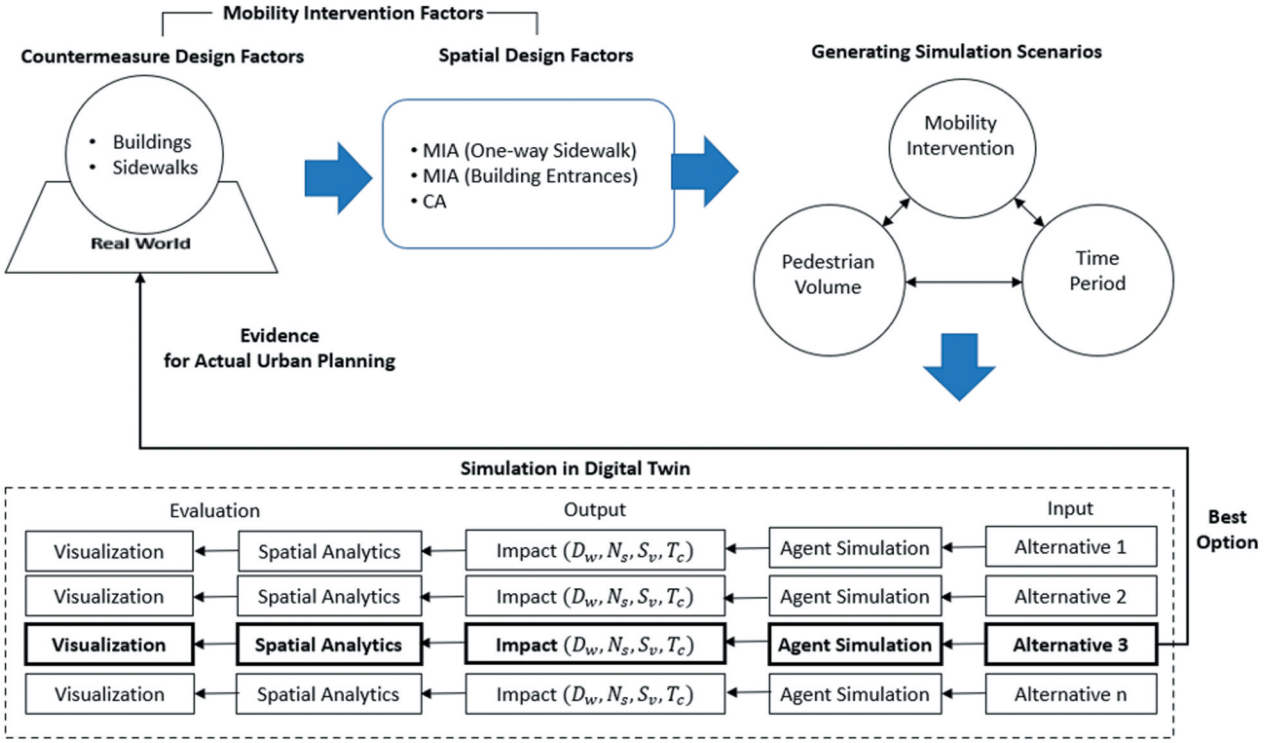


Figure 7. Illustrating a schematic diagram of experimental design for generating simulation scenarios incorporating intervention designs and assessing the effects of pedestrian movements on the generated simulations.

The experiment begins by choosing a region of interest within a campus setting, with the goal of enhancing both the effectiveness and safety of pedestrian movement. This research considers two countermeasure design factors: managing building entrances (open or closed) and altering the flow of pedestrian movement (unidirectional or bi-directional movement). The region of interest chosen is segmented into three distinct sub-regions: an area with one-sided sidewalks, an area with controlled building entrances, and the connecting regions between the two. These countermeasures and spatial design factors serve as mobility intervention components in Figure 7. Additionally, the number of pedestrian agents is generated at two levels (basic and doubled volume). Considering different movement patterns based on the time of day (morning and daytime), the simulation can produce multiple alternate scenarios for pedestrian movement by combining mobility interventions and pedestrian volume. The developed digital twin system quantitatively evaluates the effects of all possible scenarios with various measures concerning the movement efficiency, physical spacing and close contact time. These impact measures were visualized and analyzed through a spatial information system to determine the optimal decision, which can then be implemented in real-world applications.

4.2. Variables description

The experiment aimed to examine the effects of mobility interventions, such as one-way sidewalks and building entrance controls, on the pedestrian spacing of pedestrians in campus walking areas. A common variable for pedestrian spacing used in several related studies (Harweg, Bachmann, and Weichert 2021; Romero, Stone, and Ford 2020) was employed.

The close contact time for each agent is within a 2-m distance of other agents. Factoring in the number of close contacts, a 1-s contact with a single agent within 2 m would count as 1 s of close contact time, while contact with two other agents for 1 s each would count as 2 s. This variable was aggregated for each agent and used in the analysis. The average close contact time within 2 m of other agents (in seconds) was calculated using Equation (1):

$$T_c = \frac{\sum_i \left(\sum_k k \cdot t_k\right)_i}{n} \quad (1)$$

where k represents the number of agents contacted within 2-m distance, t_k represents contacted time within 2 m with k agents, i is each individual agent, and n is the total number of agents.

In this research, enhanced pedestrian spacing was sought by dispersing pedestrians through mobility interventions. To gauge pedestrian dispersion, two variables were employed: the number of sidewalk segments utilized by agents and the standard deviation of pedestrian volume for each segment. These variables indicate the degree of pedestrian dispersion. The calculation for the number of used segments is presented in Equation (2):

$$N_s = \sum_j S_j \tag{2}$$

where S_j is set to 0 if no agents use the j-th segment, and it is set to 1 if one or more agents use the j-th segment. The standard deviation of pedestrian volume across segments is computed using Equation (3):

$$S_v = \sqrt{\frac{\sum_j (V_j - V)^2}{N}} \tag{3}$$

where V_j represents the number of agents using j-th segment, denotes the average number of agents using the segments, and N indicates the total number of segments. As the number of utilized sidewalk segments increases and the standard deviation of the pedestrian volume for each segment decreases, agent dispersion becomes more pronounced.

Lastly, to assess the potential detour effect of pedestrian dispersion, the average walking distance for agents was measured. This value is calculated using Equation (4):

$$D_w = \frac{\sum_i d_i}{n} \tag{4}$$

where d_i represents the walking distance of agent i, and n denotes the total number of agents.

4.3. Scenario setting

For setting up the simulation scenario, we utilized CCTV footage. To implement pedestrian flow in the actual campus space, we checked the pedestrian volume and movement directions at major spots on campus and reflected them in the scenario settings. A range of simulation scenarios was analyzed to determine how the effects of mobility interventions vary under different circumstances. As illustrated in Figure 8, agents' movement flows were categorized into two types. Figure 8(a) displays the movement flow of pedestrians during the morning hours, representing the time when university members arrive at campus from their residences. In this scenario, agents move from the subway station (Building No. 1) to eight other buildings ($1 \times 8 = 8$ Origin-Destination (OD)s). Based on baseline campus pedestrian-volume data derived from CCTV footage, agents were configured to randomly spawn at a rate of 1 agent per 30 seconds from the subway station to each destination. Figure 8(b) depicts the movement flow of campus pedestrians during daytime hours, representing a situation where pedestrian flow occurs between all buildings ($9 \times 8 = 72$ ODs). In this case, agents were set to randomly spawn at a rate of 1 agent per 150 s from each origin to each destination, also based on campus walking volume data.

Mobility interventions were designed, considering the unique features of the campus sidewalk network. Figure 9(a) displays the campus sidewalk network without any interventions, with letters denoting the entrances of each building. Figure 9(b) presents the network with mobility interventions applied. First, one-way sidewalks were implemented along the paths from the subway station (Building No.1) to the campus center, as indicated by the arrows. This area was deemed suitable for one-way sidewalks due to the proximity of parallel sidewalks and the absence of specific destinations along the route. Entrance controls were applied to Buildings 3 to 8, which are the primary buildings at the campus center. Entrances marked with an "X" in the figure are assumed to be closed. The Mobility Intervention Area (MIA) can be observed in Figure 9(b), where each intervention is applied. An additional Connecting Area (CA) is defined as the area located between the two MIAs, even though neither of the two interventions is applied there.

The study adapted the application of mobility interventions for each time scenario based on feedback from initial simulation results. However, when both one-way sidewalks and building entrance control were applied in simulation,

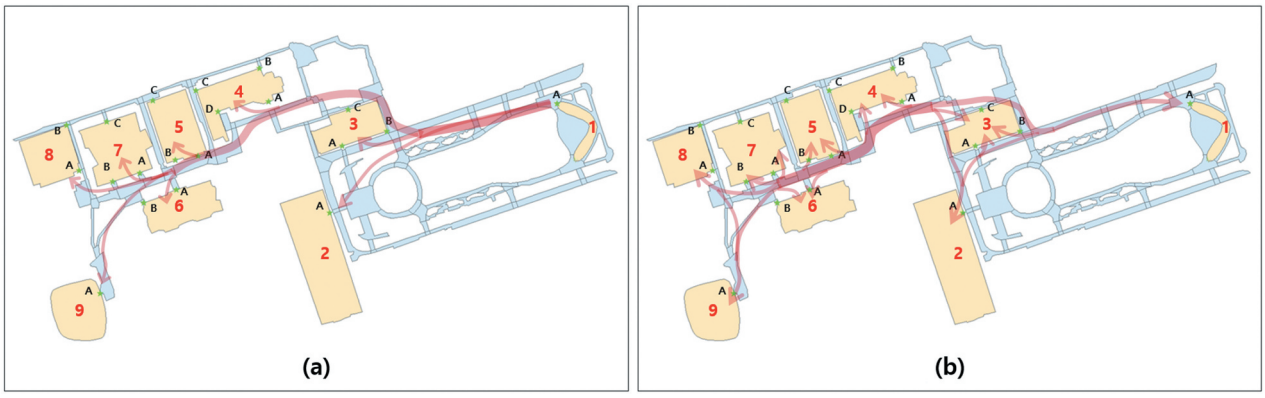


Figure 8. Two types of the flow of the agents' movements: (a) Morning time scenario: one-way flow from the subway station to the campus center, and (b) Daytime scenario: random flow among all buildings.

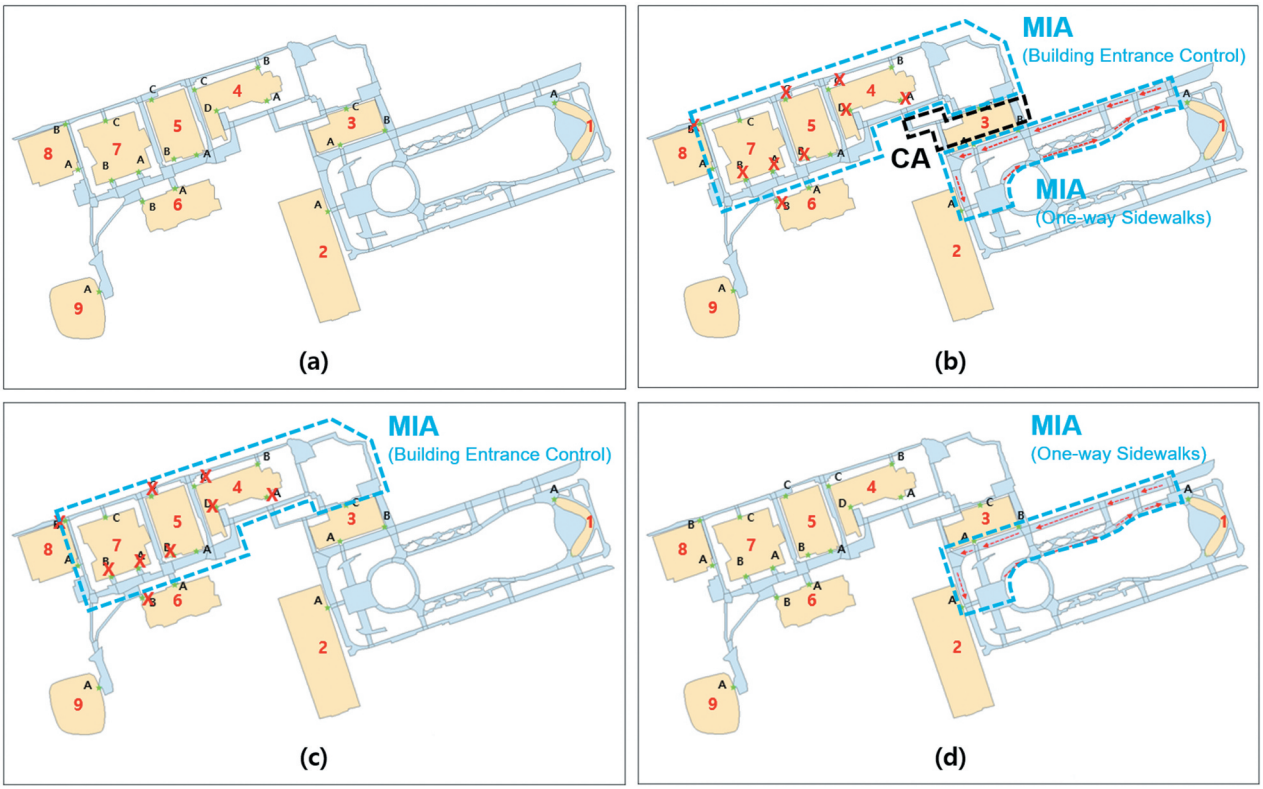


Figure 9. Implementing mobility interventions on the campus sidewalk network: (a) Campus sidewalk network without mobility interventions, (b) Campus sidewalk network with one-way sidewalk and building entrance control applied (for initial simulation), (c) Campus sidewalk network with only building entrance control applied (morning time scenario), and (d) Campus sidewalk network with only one-way sidewalk applied (daytime scenario).

pedestrian spacing deteriorated in both morning and daytime scenarios. In the morning time scenario, physical distancing worsened on the sidewalks between the subway station and the campus center, as agents who used parallel sidewalks became more concentrated due to the one-way sidewalks. This disruption of agents' dispersion confirmed that one-way settings on sidewalks can be counterproductive if the agents' movement is already one-way. In the daytime scenario, pedestrian spacing worsened on the sidewalks in the campus center due to the significant increase in agents' walking distance caused by the building entrance control. As a result, it was recommended to apply only building entrance control (Figure 9(c)) in the morning when the agents' movements are one-way and only one-way sidewalks (Figure 9(d)) in the daytime when the agents' movements are random to improve pedestrian spacing through agents' dispersion.

Pedestrian volume was identified as a critical factor affecting pedestrian spacing, with increasing volume leading to higher pedestrian density, reduced space available for each pedestrian, and increased close contacts between pedestrians. To evaluate the effect of mobility interventions on pedestrian volume, additional simulation scenarios were conducted with twice the pedestrian volume.

Overall, the study conducted eight scenarios based on three criteria: time period, application state of the mobility intervention, and pedestrian volume, as summarized in Table 1.

Table 1. Scenarios for pedestrian simulation.

Scenario	Time period	Mobility intervention	Pedestrian volume
S1(M-N-B)	Morning	No intervention	Basic
S2(M-A-B)	Morning	Applied (building entrance control)	Basic
S3(D-N-B)	Daytime	No intervention	Basic
S4(D-A-B)	Daytime	Applied (one-way sidewalk)	Basic
S5(M-N-D)	Morning	No intervention	Doubled
S6(M-A-D)	Morning	Applied (building entrance control)	Doubled
S7(D-N-D)	Daytime	No intervention	Doubled
S8(D-A-D)	Daytime	Applied (one-way sidewalk)	Doubled

Note: S1–S8 represents eight different scenarios creating by systematically combining several design factors; "M"/"D" represents a scenario considering the pedestrian moving patterns shown in morning time or daytime respectively; "A"/"N" indicates where the mobility intervention (building entrance control or one-way sidewalk) is applied ("A") or not ("N"); "Basic"/"Doubled" means the two categorical volumes of pedestrian agents generated.

5. Experimental results and discussion

5.1. Experimental setting

By default, the simulator establishes whether to implement interventions, simulation duration, agent spawning rate per OD, and agent spawning interval. Interventions are applied based on the predefined scenarios in the experimental design. All scenarios have a simulation time of 15 min. Agents are spawned at random intervals according to the specified flow rate, with the choice between random and uniform intervals; random is more suitable as it reflects real-world irregular movement patterns.

Agent spawning flow rates per OD vary depending on the scenario. For morning scenarios, agents are assumed to move from the subway station to each building, forming an OD matrix with 8 ODs. Pedestrian volume is based on actual campus pedestrian traffic, obtained from CCTV footage at York University. By gradually increasing the agent spawning rate for each simulated OD, we matched the agent volume in the digital twin pathways with the actual video data, ultimately setting a default flow rate of 30 agents per 15 min for each of the 8 ODs. For daytime scenarios, the flow rate is similarly determined, assuming movement between all buildings and creating an OD matrix with 72 ODs. For each of the 72 ODs, the basic flow rate is set to spawn 6 agents per 15 min. In congested scenarios (5 through 8), the agent spawning flow rate is doubled.

We conducted simulations with these settings, running each of the 8 scenarios three times. Due to the simulation's nature, agents begin spawning immediately upon starting, resulting in fewer agents in the digital twin space initially. After 15 min, the number of agents decreases as no new agents' spawn. To obtain accurate results for close contacts between agents, we analyzed records between 5 min and 15 min after the simulation's start.

On average, in the morning scenarios S1 and S2, 67 and 71 agents simultaneously walked through the digital twin campus, respectively. In congested scenarios S5 and S6, the average number of agents walking simultaneously was 129 and 140, respectively. In daytime scenarios S3 and S4, the average number of simultaneous agents was 65 and 67, respectively. In congested scenarios S7 and S8, the average number of simultaneous agents was 145 and 144, respectively. We used the position data extracted from these simulations to analyze pedestrian spacing between agents.

5.2. Results

In all scenarios, we observed that the intervention led to an increase in the agents' travel distance while simultaneously reducing their close contact time with other agents. This pattern held true for both morning and daytime scenarios. The intervention's effectiveness remained consistent even as the increase in pedestrian volume inherently led to more close contact time between agents.

Before implementing the intervention in S1, the agents had an average travel distance of 406 m used 46 segments, had a standard deviation of pedestrian volume per segment of 74.6, and experienced 74.1 s of close contact time per agent. After the intervention was introduced, resulting in S2, these metrics shifted to 421 m, 62 segments, a 66.3 standard deviation, and 62.2 s of close contact time per agent. The intervention led to a minor increase in travel distance while improving the dispersion within the campus area, ultimately reducing close contact time.

In S3, prior to the intervention, the agents' average travel distance was 233 m, with 81 segments used, a 54.1 standard deviation of pedestrian volume, and 26.5 s of close contact time per agent. When the intervention was applied, creating S4, these values changed to 250 m, 80 segments, a 51.4 standard deviation, and 21.3 s of close contact time per agent. Similar to the morning scenario, the daytime intervention led to a slight increase in travel distance and improved dispersion in the campus space, resulting in reduced close contact time.

Before the intervention in S5, with twice the basic morning pedestrian volume, the agents' average travel distance was 407 m, with 55 segments used, a 146.6 standard deviation of pedestrian volume per segment, and 115.0 s of close contact time per agent. After implementing the intervention, forming S6, these values shifted to 421 m, 62 segments, a 130.0 standard deviation, and 101.1 s of close contact time per agent. Once again, the intervention led to a minor increase in travel distance and improved dispersion in the campus area, resulting in reduced close contact time.

In S7, with double the daytime basic pedestrian volume, the agents' average travel distance was 232 m, 82 segments were used, the standard deviation of pedestrian volume per segment was 114.5, and the close contact time per agent was 56.1 s. After applying the intervention, creating S8, these metrics changed to 253 m, 81 segments, a 107.2 standard deviation, and 48.1 s of close contact time per agent. Similar to the other scenarios, even with doubled pedestrian volume during daytime, the intervention slightly increased agents' travel distance and improved the dispersion within the campus space, leading to a reduction in close contact time.

5.3. Discussion

Table 2 presents the simulation results comparing the effect of the building entrance control on pedestrian spacing in the morning time (S1 (M-N-B)) and with the control in effect (S2 (M-A-B)). The average

Table 2. Simulation results for the defined scenarios.

		Pedestrian dispersion			
Scenario	D_w (m)	N_s	S_v	T_c (s)	Change in T_c
S1(M-N-B)	406	46	74.6	74.1	–
S2(M-A-B)	421	62	66.3	62.2	16% reduction (S1–S2)
S3(D-N-B)	233	81	54.1	26.5	–
S4(D-A-B)	250	80	51.4	21.3	19% reduction (S3–S4)
S5(M-N-D)	407	55	146.3	115.0	–
S6(M-A-D)	421	62	130.0	101.1	12% reduction (S5–S6)
S7(D-N-D)	232	82	114.5	56.1	–
S8(D-A-D)	253	81	107.2	48.1	14% reduction (S7–S8)

Note: Dw, Ns, Sv, and Tc are defined in Equations (1)–(4) in Section 4.1. These variables represent respectively: the average walking distance for all agents, the number of sidewalk segments utilized by agents, the standard deviation of the number of pedestrians occupied on segments, and average time in close contact.

walking distance of agents increased by 4% due to the building entrance control, but the number of sidewalk segments used by agents increased, and the standard deviation of the number of agents on a segment decreased by 11%, resulting in a 16% reduction in agents' average time in close contact within 2 m with other agents. This confirms the improvement of pedestrian spacing between agents in the sidewalk space. A similar effect was found with one-way sidewalks in the daytime scenario (S3 (D-N-B) vs. S4 (D-A-B)), with the average walking distance of agents increasing by 7% but the standard deviation of the number of agents on a segment decreasing by 5%, resulting in a 19% reduction in agents' average time in close contact within 2 m with other agents.

S5 to S8 in Table 2 were scenarios where the pedestrian volume was doubled from S1 to S4, and it was found that the mobility interventions helped improve the pedestrian spacing even with the increased volume. In S6 (M-A-D) and S8 (D-A-D), the agents' walking distance increased, and agents were dispersed in the sidewalk network, resulting in a decrease in the agents' average close contact time within 2 m with other agents. However, the effect of the mobility intervention on improving pedestrian spacing decreased as the pedestrian volume increased. The average close contact time in S6 (M-A-D) decreased by 12% compared to S5 (M-N-D), and in S8 (D-A-D) decreased by 14% compared to S7 (D-N-D), which is a significant decrease in effect compared to the basic pedestrian volume (16% and 19%). This weakening of the mobility intervention's effect on improving pedestrian spacing in situations with a high pedestrian volume is expected to be due to a large number of close contacts between pedestrians occurring even on distributed sidewalks.

The simulation recorded all instances of 2 m distancing violations, and the spatial pattern of close contacts was analyzed based on this data. Figure 10 illustrates the morning time scenarios, and Figure 11 shows the daytime scenarios. The analysis revealed that the mobility interventions had a more significant effect in the vicinity of the MIA (middle of the campus in the morning

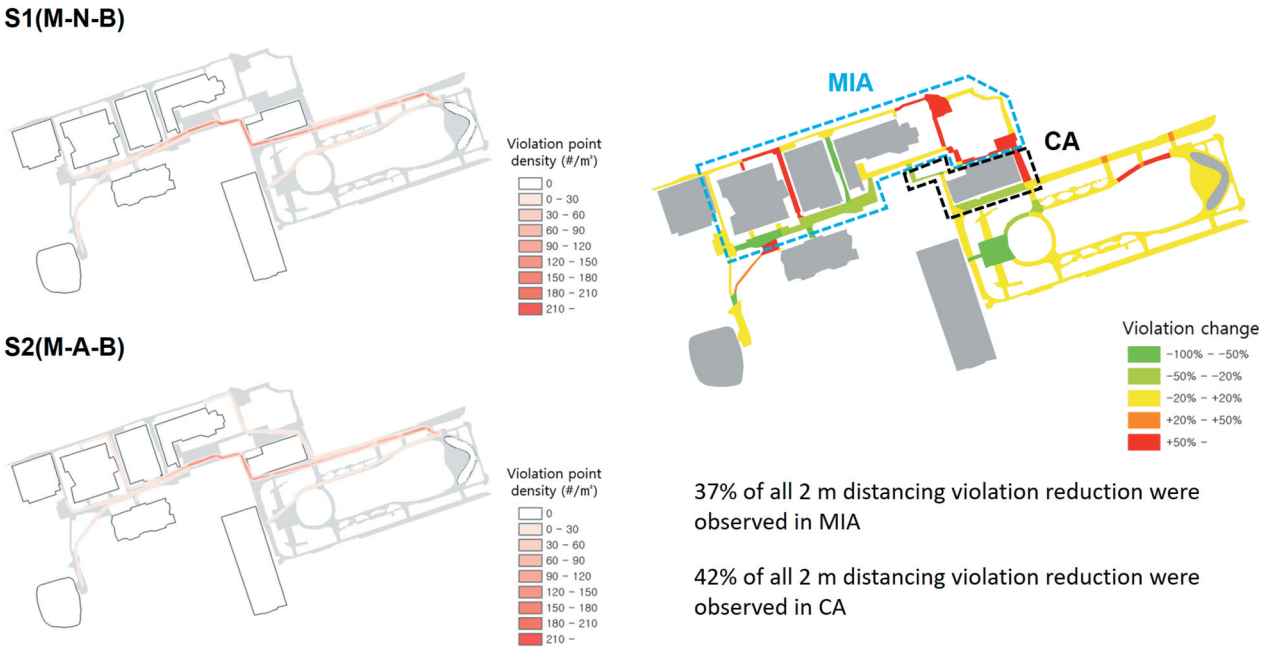


Figure 10. Spatial analysis of morning time scenarios (S1 vs. S2): effect of building entrance control on pedestrian close contact (S1 = No control; S2 = control in effect).

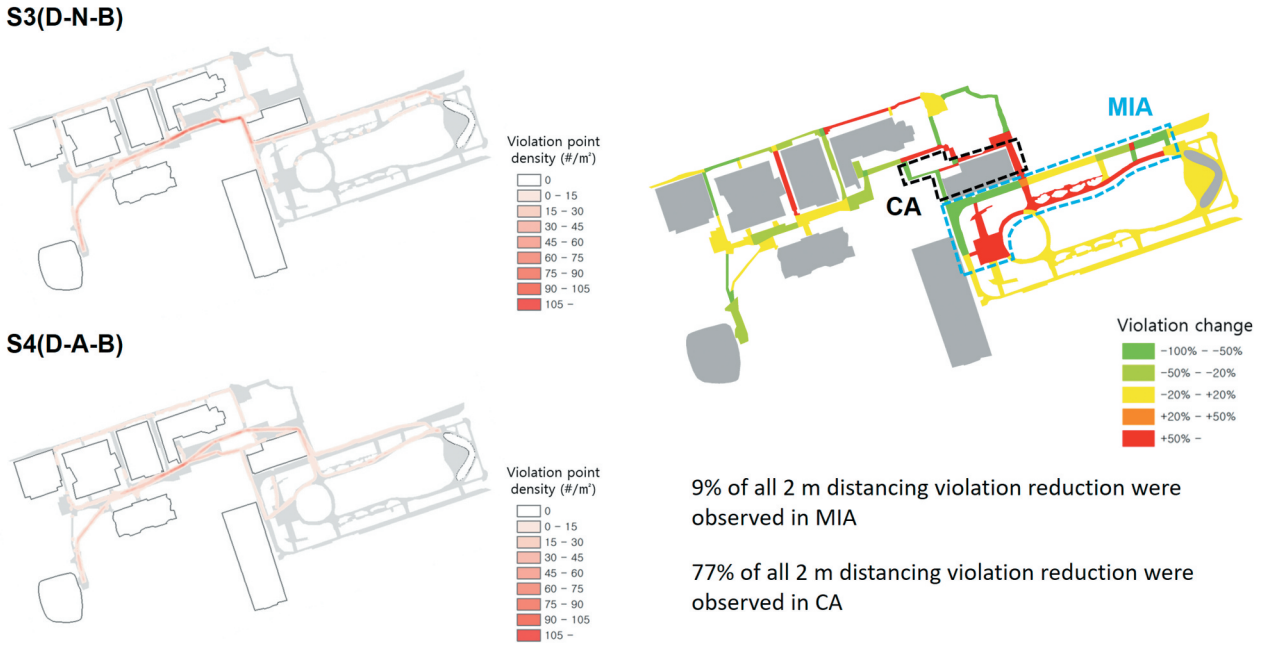


Figure 11. Spatial analysis of daytime scenarios (S3 vs. S4): effect of one-way sidewalk on pedestrian close contact (S3 = No control; S4 = control in effect).

scenarios and sidewalks between the subway station and the campus center in the daytime scenarios) rather than the area where the interventions were applied. Specifically, there was a noticeable reduction in close contacts in the CA between the campus center and the sidewalks connecting to the subway station.

The comparison of S2 (M-A-B) and S1 (M-N-B) revealed that 37% of the total reduction in 2 m distancing violations occurred in MIA, and 42% occurred in CA. Similarly, the comparison of S4 (D-A-B) and S3 (D-N-B) showed that only 9% of the total reduction in close contacts occurred in MIA, and 77% occurred in CA. CA is an area that connects the region near the subway station and the center of the campus, where a bottleneck arises due to narrow sidewalks. The application of mobility interventions resulted in a dispersion effect on pedestrians, leading to a more substantial impact not only in the intervention area but also in the adjacent bottleneck area (CA). This pattern was observed in all comparative analyses (S1 vs. S2, S3 vs. S4, S5 vs. S6, and S7 vs. S8), irrespective of pedestrian flow volume.

As a result, we could confirm that accessible entrances or sidewalks for pedestrians could become a major environmental factor affecting pedestrian spacing in our experiment. Pedestrian dynamic variables such as pedestrian flow and volume set based on CCTV data were also analyzed to affect the experimental results. However, our experiment could demonstrate whether the intended effect could be achieved through small changes to the physical environment of an existing city from the perspective of tactical urbanism.

6. Conclusions

In this study, we present a novel simulation in digital twin by combining evidence-based planning and tactical urbanism, with temporary mobility interventions proposed as a specific approach to enhancing pedestrian spacing in a university campus under high-density pedestrian conditions. The effectiveness of these interventions has been demonstrated through agent-based simulations, yielding the following conclusions based on the experimental results.

First, this study developed a digital twin system for a real university campus space to simulate pedestrian movement in urban settings. Rather than relying solely on GIS data for 3D modeling, the digital twin concept was implemented by incorporating agent-based simulation functionality into the campus model. Although real-time pedestrian data was not used in the simulation, the urban digital twin's utility was demonstrated by generating meaningful results through various scenario experiments.

Second, this study proposed building entrance control and one-way sidewalks as mobility interventions for promoting pedestrian spacing on sidewalk. These relatively simple, low-cost measures can be temporarily implemented with ease. The simulation results indicate that they can effectively improve pedestrian spacing by dispersing pedestrians from congested to under-utilized sidewalks.

Third, the refinement of these mobility interventions requires simulation feedback. Experiments on multiple scenarios confirmed that the impact of such interventions can vary depending on factors like

pedestrian flow patterns and volumes. Applying a specific intervention without careful consideration may produce undesirable side effects. The study found that effective interventions differ based on pedestrian flow patterns (one-way in the morning, random in the daytime), and the spatial pattern revealing their impact was more complex than initially assumed. The interventions' effectiveness also depended on pedestrian volume, with higher volumes reducing their efficacy. In high-traffic areas, more potent alternatives such as spatial expansion or demand restriction may be necessary. Consequently, planners must thoroughly analyze the pedestrian spacing conditions in the target area and respond cautiously. Evidence-based, flexible decision-making is required to determine the applicability, type, method, and intensity of interventions. Effective cross-sectoral cooperation demands scientific evidence to support decision-making. Our developed system can swiftly provide such evidence in an interdisciplinary context.

Fourth, while the study's findings cannot be directly generalized to real-world settings due to its limitations, they nonetheless hold significance by demonstrating potential applicability in analogous contexts. The research employed a simulation-based experiment on a university campus with relatively straightforward pedestrian patterns, whereas actual urban spaces and pedestrian movements are more complex. However, this study does confirm the potential for proposed mobility interventions to be flexibly utilized for improving pedestrian spacing in various situations. For instance, during severe pandemic periods, mobility interventions proposed in this study could be applied to outdoor public facilities such as parks. Unlike the destination-oriented pedestrian flows commonly observed in campus environments, park visitors tend to move more freely, rendering one-way sidewalk systems suboptimal. However, by analyzing park entry and exit patterns, temporary measures such as establishing additional access points could be implemented to anticipate and mitigate crowd density. Similarly, in high-density residential complexes comprising multiple buildings, controlling primary building entrances could potentially prevent overconcentration at specific locations.

Future research should focus on providing more robust scientific evidence to support planners' decision-making. In particular, sensitivity analyses should examine proximity thresholds other than the 2 m baseline so that different risk levels and design standards can be evaluated. Data-driven urban planning theories, such as space syntax (Hillier and Hanson 1989), can be employed to connect sidewalk network characteristics with intervention applications. In terms of improving the quality of the simulation, it is expected that the realism of pedestrian simulations can be enhanced through the analysis of CCTV footage of pedestrian behaviors under high-density pedestrian environments or via VR co-simulation techniques, and by incorporating large-scale mobility datasets – such as anonymized mobile-phone location data and Wi-Fi /Bluetooth probe data – into the digital-twin setup for calibration and validation. Additionally, artificial intelligence technology can be used to derive optimal application methods based on specific circumstances.

Author contributions

CRediT: **Seungho Yang:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Muhammad Usman:** Data curation, Funding acquisition, Software, Visualization, Writing – original draft; **Ali Hassan:** Data curation, Software; **Gunho Sohn:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Visualization, Writing – original draft, Writing – review & editing; **Petros Faloutsos:** Supervision; **Brandon Haworth:** Data curation, Software.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Hanbat National University Research Fund (2021); the Faculty of Science start-up funds at Ontario Tech University; and the Intelligent Systems for Sustainable Urban Mobility (ISSUM) project.

Notes on contributors

Seungho Yang received his B.S. in civil engineering (2009) and the PhD in urban design (2014) from Seoul National University. He was a Postdoctoral Researcher at York University (2018–2020). Since 2021, he has been with the Department of Urban Engineering, Hanbat National University, where he is currently an Associate Professor. His research interests include smart cities, digital twins, and urban design/analytics.

Muhammad Usman received his PhD in computer science from York University in 2020. Following his PhD, he was a Postdoctoral Fellow with the Ontario Research Fund/ Intelligent Systems for Sustainable Urban Mobility (ORF/ ISSUM) project. He also served as a senior postdoctoral researcher in the Intelligent Visual Interfaces Lab at Rutgers University. He later served as an Assistant Professor in the Department of Information and Computer Science at King Fahd University of Petroleum and Minerals. He is currently an Assistant Professor in the Faculty of Science in the Computer Science group, and the Director of Graphics and Virtual Dynamics (GVD) Lab at Ontario Tech University. His research lies at the intersection of crowd simulation, behavioral modeling, human-building interaction, generative AI for games, and emotionally-aware virtual environments.

Ali Hassan is a visualization analyst at Candu Energy Incorporated. He received his M.Sc. in earth and space science from York University (2023) and his B.Sc. in computer science from the University of the Punjab (2013). His research and professional interests focus on digital twin development, high-fidelity 3D and technical visualization, simulation of autonomous robotic systems, process automation and PLM integration, and human-centered interactive visual environments for complex engineering and industrial applications.

Gunho Sohn received the B.Sc. and M.Sc. in astronomy and space science from Yonsei University and the PhD. in geomatics engineering from University College London (2004). He is a Professor in the Department of Earth and Space Science and Engineering and the Inaugural Director of the Mobility Innovation Centre, Lassonde School of Engineering, York University. His research focuses on spatial data analytics, computer vision, and machine learning for large-scale urban digital twins, with extensive industry-collaborative projects funded by CFI, NSERC, and the Ontario Research Fund. He has authored 150+ peer-reviewed publications and received the 2004 Best Thesis Award (British Remote Sensing and Photogrammetry Society) and York Research Leadership Awards.

Petros Faloutsos is a Professor in the Department of Electrical Engineering and Computer Science at York University, and an Affiliate Scientist at the UHN-Toronto Rehabilitation Institute. Before joining York, he was a faculty member in the Computer Science Department at the University of California at Los Angeles, where in 2002, he founded the first computer graphics lab at UCLA. Faloutsos received his PhD degree (2002) and his M.Sc. degree (1995) in computer science from the University of Toronto and his BEng degree (1993) in electrical engineering from the National Technical University of Athens. His research interests span computer graphics, autonomous agents for games and virtual environments, motion control and facial animation, as well as interactive graphics, and real-time photorealistic rendering.

Brandon Haworth received the PhD in 2019 from York University and was a Post-Doctoral Fellow with the Ontario Research Fund/Intelligent Systems for Sustainable Urban Mobility (ORF/ISSUM) project in 2019/2020. He is an Associate Professor in the Department of Computer Science, Faculty of Engineering & Computer Science, the Director of the Graphics, Artificial Intelligence, Design, & Games (GAIDG) Lab, and a Research Fellow at the Institute on Aging and Lifelong Health at the University of Victoria. His research interests include graphics, agent-based modelling and crowd simulation, multi-agent reinforcement learning and deep learning, as well as human–computer interaction, human-centred design, artificial intelligence, and augmented intelligence.

ORCID

Seungho Yang http://orcid.org/0000-0001-8371-599X
Muhammad Usman http://orcid.org/0000-0003-2059-7206
Ali Hassan http://orcid.org/0009-0002-3064-2349
Gunho Sohn http://orcid.org/0000-0002-5127-8358
Petros Faloutsos http://orcid.org/0000-0002-4508-010X
Brandon Haworth http://orcid.org/0000-0001-8134-0047

Data availability statement

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

References

Alonso, L., Y. Zhang, A. Grignard, A. Noyman, Y. Sakai, M. ElKatsha, R. Doorley, and K. Larson. 2018. "Cityscope: A Data-Driven Interactive Simulation Tool for Urban Design. Use Case VOLPE." In *International conference on complex systems*, Cambridge, MA, USA. July 22–27.

Andres, L. 2013. "Differential Spaces, Power Hierarchy and Collaborative Planning: A Critique of the Role of Temporary Uses in Shaping and Making Places." *Urban Studies* 50 (4): 759–775. https://doi.org/10.1177/0042098012455719.

Angelidou, M., A. Psaltoglou, N. Komninos, C. Kakderi, P. Tsarchopoulos, and A. Panori. 2017. "Enhancing Sustainable Urban Development Through Smart City Applications." *Journal of Science and Technology Policy Management* 9 (2): 146–169. https://doi.org/10.1108/JSTPM-05-2017-0016.

Asadi-Shekari, Z., M. Moeinaddini, and M. Shah. 2014. "A Pedestrian Level of Service Method for Evaluating and Promoting Walking Facilities on Campus Streets." *Land Use Policy* 38:175–193. https://doi.org/10.1016/j.landusepol.2013.11.007.

Barata, A., and A. Fontes. 2017. "Tactical Urbanism and Sustainability: Tactical Experiences in the Promotion of Active Transportation." *International Journal of Urban and Civil Engineering* 11 (6): 734–739. https://doi.org/10.5281/zenodo.1130703.

Batty, M. 2018. "Digital Twins." *Environment and Planning B: Urban Analytics and City Science* 45 (5): 817–820. https://doi.org/10.1177/2399808318796416.

Cardiff News Room. 2020. "Cardiff 's Plans to Exit Lockdown as One of UK's 'Safest' Cities Revealed." Accessed February 18, 2022. https://www.cardiffnewsroom.co.uk/releases/c25/24027.html.

Champion, A. G. 2001. "A Changing Demographic Regime and Evolving Polycentric Urban Regions: Consequences for the Size, Composition and Distribution of City Populations." *Urban Studies* 38 (4): 657–677. https://www.jstor.org/stable/43198204.

Cheshmehzangi, A. 2016. "Multi-Spatial Environmental Performance Evaluation Towards Integrated Urban Design: A Procedural Approach With Computational Simulations." *Journal of Cleaner Production* 139:1085–1093. https://doi.org/10.1016/j.jclepro.2016.08.151.

Chu, D. K., E. A. Akl, S. Duda, K. Solo, S. Yaacoub, and H. J. Schünemann. 2020. "Physical Distancing, Face Masks, and Eye Protection to Prevent Person-to-Person Transmission of SARS-CoV-2 and COVID-19: A Systematic Review and Meta-Analysis." *The Lancet* 395 (10242): 1973–1987. https://doi.org/10.1016/S0140-6736(20)31142-9.

Chung, M., and M. Choo. 2011. "Computational Fluid Dynamics for Urban Design: The Prospects for Greater Integration." *International Journal of Architectural Computing* 9 (1): 33–53. https://doi.org/10.1260/1478-0771.9.1.33.

City of Westmount. 2020. "Walking in the Same Direction to Save Lives (City of Westmount." Accessed February 18,

2022. https://westmount.org/en/covid-19-walking-in-the-same-direction-to-save-lives/ .
Davis, J., P. Lin, A. Borning, B. Friedman, P. Kahn, and P. Waddell. 2006. "Simulations for Urban Planning: Designing for Human Values." *Computer* 39 (9): 66–72. https://doi.org/10.1109/MC.2006.324.
Deren, L., Y. Wenbo, and S. Zhenfeng. 2021. "Smart City Based on Digital Twins." *Computational Urban Science* 1 (1): 4. https://doi.org/10.1007/s43762-021-00005-y.
Dufour, O., H. Dang, J. Cordes, R. Korbmacher, M. Chraibi, B. Gaudou, A. Nicolas, and A. Tordeux. 2025. "Dense Crowd Dynamics and Pedestrian Trajectories: A Multiscale Field Dataset from the Festival of Lights in Lyon." *Scientific Data* 12 (1): 718. https://doi.org/10.1038/s41597-025-04732-3.
Ferreri, M. 2015. "The Seductions of Temporary Urbanism." *Ephemera* 15 (1): 181. https://doi.org/10.1017/9789048535828.
Gehl, J. 2013. *Cities for People*. Washington DC: Island press.
Graziano, T. 2021. "Smart Technologies, Back-to-the-Village Rhetoric, and Tactical Urbanism: Post-COVID Planning Scenarios in Italy." *International Journal of e-Planning Research* 10 (2): 80–93. https://doi.org/10.4018/IJEPR.20210401.oa7.
Gregg, K., P. Hess, J. Brody, and A. James. 2022. "North American Street Design for the Coronavirus Pandemic: A Typology of Emerging Interventions." *Journal of Urbanism: International Research on Placemaking and Urban Sustainability* 17 (4): 1–19. https://doi.org/10.1080/17549175.2022.2071970.
Harweg, T., D. Bachmann, and F. Weichert. 2021. "Agent-Based Simulation of Pedestrian Dynamics for Exposure Time Estimation in Epidemic Risk Assessment." *Journal of Public Health* 31:221–228. https://doi.org/10.1007/s10389-021-01489-y.
Hassan, A., M. Usman, M. Kremer, S. Yang, M. Luubert, P. Faloutsos, G. Hall, and G. Sohn. 2023. "Digital Twins: Simulating Robot-Human Sidewalk Interactions." In *2023 International 3D Geoinfo Conference*, Munich, Germany, September 12-14.
Hillier, B., and J. Hanson. 1989. *The Social Logic of Space*. Cambridge: Cambridge university press.
Jeddoub, I., G. Nys, R. Hajji, and R. Billen. 2023. "Digital Twins for Cities: Analyzing the Gap Between Concepts and Current Implementations with a Specific Focus on Data Integration." *International Journal of Applied Earth Observation and Geoinformation* 122:103440. https://doi.org/10.1016/j.jag.2023.103440.
Jeong, D., W. Choi, S. Hong, K. Cho, and J. Park. 2023. "Comparative Analysis of Digital Twin Trends in Urban Area." *Journal of Cadastre & Land Informatix* 53 (1): 123–137. https://doi.org/10.22640/lxsiri.2023.53.1.123.
Kang, B., A. Moudon, P. Hurvitz, and B. Saelens. 2017. "Differences in Behavior, Time, Location, and Built Environment Between Objectively Measured Utilitarian and Recreational Walking." *Transportation Research Part D: Transport and Environment* 57:185–194. https://doi.org/10.1016/j.trd.2017.09.026.
Kapadia, M., N. Pelechano, J. Allbeck, and N. Badler. 2015. "Virtual Crowds: Steps Toward Behavioral Realism." *Synthesis Lectures on Visual Computing: Computer Graphics, Animation, Computational Photography, and Imaging* 7 (4): 1–270. https://doi.org/10.2200/S00673ED1V01Y201509CGR020.
Karamouzas, I., P. Heil, P. van Beek, and M. Overmars. 2009. "A Predictive Collision Avoidance Model for Pedestrian Simulation." In *International workshop on motion in games*, Zeist, The Netherlands. November 21–24.
Karedes, D. 2020. Beverly Police Enforcing One-Way Sidewalks in Effort to Promote Social Distancing. Boston 25 News, 2020. Accessed February 18, 2022. https://westmount.org/en/covid-19-walking-in-the-same-direction-to-save-lives.
Karimi, H., B. Ghadirifaraz, S. N. Shetab Boushehri, S. Hosseininasab, and N. Rafiei. 2022. "Reducing Traffic Congestion and Increasing Sustainability in Special Urban Areas Through One-Way Traffic Reconfiguration." *Transportation* 49:37–60. https://doi.org/10.1007/s11116-020-10162-4.
Law, L., S. Azzali, and S. Conejos. 2021. "Planning for the Temporary: Temporary Urbanism and Public Space in a Time of COVID-19." *The Town Planning Review* 92 (1): 65–73. https://doi.org/10.3828/tpr.2020.48.
Lehtola, V., M. Koeve, S. Elberink, P. Raposo, J. Virtanen, F. Vahdatikhaki, and S. Borsci. 2022. "Digital Twin of a City: Review of Technology Serving City Needs." *International Journal of Applied Earth Observations and Geoinformation* 114:102915. https://doi.org/10.1016/j.jag.2022.102915.
Li, C., J. Yin, and L. Chen. 2023. "Impact of Social Distancing on Disease Transmission Risk in the Context of a Pandemic." *Physical Review E* 108 (5): 054115. https://doi.org/10.1103/PhysRevE.108.054115.
Lydon, M., and A. Garcia. 2015. "A Tactical Urbanism How-to." In *Tactical Urbanism*, 171–208. Washington, DC: Island Press. https://doi.org/10.5822/978-1-61091-567-0_5
Madane, S., and D. Chitre. 2021. "Social Distancing Detection and Analysis through Computer Vision." In *2021 6th international conference for convergence in technology*, Mumbai, India. April 2–4.
Moonen, P., T. Defraeye, V. Dorer, B. Blocken, and J. Carmeliet. 2012. "Urban Physics: Effect of the Micro-Climate on Comfort, Health and Energy Demand." *Frontiers of Architectural Research* 1 (3): 197–228. https://doi.org/10.1016/j.foar.2012.05.002.
Mueller, C., U. Klein, and A. Hof. 2018. "An Easy-to-Use Spatial Simulation for Urban Planning in Smaller Municipalities." *Computers, Environment and Urban Systems* 71:109–119. https://doi.org/10.1016/j.compenvurbsys.2018.05.002.
NACTO. 2020. Streets for Pandemic Response and Recovery, The National Association of City Transportation Officials. Accessed May 21, 2020. https://nacto.org/publication/streets-for-pandemic-response-recovery/.
Nochta, T., L. Wan, J. Schooling, and A. Parlikad. 2021. "A Socio-Technical Perspective on Urban Analytics: The Case of City-Scale Digital Twins." *Journal of Urban Technology* 28 (1–2): 263–287. https://doi.org/10.1080/10630732.2020.1798177.
Noland, R., D. Gao, E. Gonzales, and C. Brown. 2015. "Costs and Benefits of a Road Diet Conversion." *Case Studies on Transport Policy* 3 (4): 449–458. https://doi.org/10.1016/j.cstp.2015.09.002.
Ohtani, Y., S. Sawada, and S. Nogami. 2010. "Study on Congestion Reducing in the Theme Park." In *8th Asia-Pacific Symposium on Information and*

Telecommunication Technologies, Kuching, Malaysia, 15–18 June.

Oxford City Council. 2020. Oxford City Council Introduces One-Way Pavements and a Raft of Other Measures to Help People Maintain Social Distancing as Shops Re-Open. Oxford City Council. Accessed February 18, 2022. https://www.oxford.gov.uk/news/article/1459/oxford_city_council_introduces_oneway_pavements_and_a_raft_of_other_measures_to_help_people_maintain_social_distancing_as_shops_reopen.

Pari, N., B. Vasu, and A. Geetha. 2020. "Monitoring Social Distancing by Smart Phone App in the Effect of COVID-19." *Global Journal of Computer Science and Technology* 9 (9): 946–953. https://doi.org/10.17577/IJERTV9IS090469.

Pearson, H. 2020. Calgary's Physical Distancing-Focused Lane Closures End Oct. 13, Global News. October 9. Accessed February 18, 2022. https://globalnews.ca/news/7389758/coronavirus-calgary-physical-distancing-road-closures-end/.

Pelechano, N., J. Allbeck, M. Kapadia, and N. Badler, ed. 2016. *Simulating Heterogeneous Crowds with Interactive Behaviors*. Boca Raton: CRC Press.

Pradifta, F., G. Puspitasari, I. Indratno, and F. Fadhilah. 2021. "The Application of Tactical Urbanism in Public Space on COVID-19 Transmission Prevention." *Earth and Environmental Science* 830 (1): 012087. https://doi.org/10.1088/1755-1315/830/1/012087.

Psyllidis, A., F. Duarte, R. Teeuwen, A. Salazar-Miranda, T. Benson, and A. Bozzon. 2023. "Cities and Infectious Diseases: Assessing the Exposure of Pedestrians to Virus Transmission Along City Streets." *Urban Studies* 60 (9): 1610–1628. https://doi.org/10.1177/00420980211042824.

Romero, V., W. Stone, and J. Ford. 2020. "covid-19 Indoor Exposure Levels: An Analysis of Foot Traffic Scenarios within an Academic Building." *Transportation Research Interdisciplinary Perspectives* 7:100185. https://doi.org/10.1016/j.trip.2020.100185.

Ruki, U., and A. Wulandari. 2021. "Post COVID-19 Public Space Adaption: A Case Study of Building Entrances in Jakarta." *IOP Conference Series: Earth and Environmental Science* 794 (1): 012243. https://doi.org/10.1088/1755-1315/794/1/012243.

Schaumann, D., S. Moon, M. Usman, R. Goldstein, S. Breslav, A. Khan, P. Faloutsos, and M. Kapadia. 2020. "Join: An Integrated Platform for Joint Simulation of Occupant-Building Interactions." *Architectural Science Review* 63 (3–4): 339–350. https://doi.org/10.1080/00038628.2019.1662767.

Schneider, A., E. Krueger, B. Vollenwyder, J. Thurau, and A. Elfering. 2021. "Understanding the Relations Between Crowd Density, Safety Perception and Risk-Taking Behavior on Train Station Platforms: A Case Study from Switzerland." *Transportation Research Interdisciplinary Perspectives* 10:100390. https://doi.org/10.1016/j.trip.2021.100390.

Shao, Z., G. Cheng, J. Ma, Z. Wang, J. Wang, and D. Li. 2021. "Real-Time and Accurate UAV Pedestrian Detection for Social Distancing Monitoring in COVID-19 Pandemic." *IEEE Transactions on Multimedia* 24:2069–2083. https://doi.org/10.1109/TMM.2021.3075566.

Southworth, M. 2005. "Designing the Walkable City." *Journal of Urban Planning and Development* 131 (4): 246–257. https://doi.org/10.1061/(ASCE)0733-9488(2005)131:4(246).

Still, K., M. Papalexi, Y. Fan, and D. Bamford. 2020. "Place Crowd Safety, Crowd Science? Case Studies and Application." *Journal of Place Management and Development* 13:4)385–407. https://doi.org/10.1108/JPMD-10-2019-0090.

Storeys. 2020. New Physical Distancing Idea for Toronto Proposes Making Sidewalks 'One-Way'. April 13. Accessed February 18, 2022. https://storeys.com/toronto-sidewalks-one-way-physical-distancing.

Su, J., X. He, L. Qing, T. Niu, Y. Cheng, and Y. Peng. 2021. "A Novel Social Distancing Analysis in Urban Public Space: A New Online Spatio-Temporal Trajectory Approach." *Sustainable Cities and Society* 68:02765. https://doi.org/10.1016/j.scs.2021.102765.

Taylor, N. 1998. *Urban Planning Theory Since 1945*. London, Thousand Oaks: SAGE Publications. https://doi.org/10.4135/9781446218648.

Tyrna, B., and V. Hochschild. 2010. "Modellierung Von Lokalen Überschwemmungen Nach Starkniederschlägen." In *2010 angewandte geoinformatik conference*, Salzburg, Austria. July 7–9.

Unity. 2021. "Unity Real-Time Development Platform." Accessed November 29, 2021. https://unity.com/.

Urech, P., M. Mughal, and C. Bartesaghi-Koc. 2022. "A Simulation-Based Design Framework to Iteratively Analyze and Shape Urban Landscapes Using Point Cloud Modeling." *Computers, Environment and Urban Systems* 91:101731. https://doi.org/10.1016/j.compenvurbsys.2021.101731.

Usman, M., B. Haworth, P. Faloutsos, and M. Kapadia. 2021. "Simulation-as-a-Service: Analyzing Crowd Movements in Virtual Environments." *Computer Animation and Virtual Worlds* 32 (6): e1990. https://doi.org/10.1002/cav.1990.

Usman, M., T. Lee, R. Moghe, X. Zhang, P. Faloutsos, and M. Kapadia. 2020. "A Social Distancing Index: Evaluating Navigational Policies on Human Proximity Using Crowd Simulations." In *Motion, Interaction and Games Virtual Meeting*, Virtually (online), October 16–18.

Usman, M., D. Schaumann, B. Haworth, M. Kapadia, and P. Faloutsos. 2019a. "Joint Exploration and Analysis of High-Dimensional Design–Occupancy Templates." In *Motion, Interaction and Games*, Tyne, England. Oct 28–30.

Usman, M., D. Schaumann, B. Haworth, M. Kapadia, and P. Faloutsos. 2019b. "Joint Parametric Modeling of Buildings and Crowds for Human-Centric Simulation and Analysis." In *International Conference on Computer-Aided Architectural Design Futures*, Daejeon, Korea. June 26–28.

Walliss, J. 2018. "Landscape Architecture and the Digital Turn: Towards a Productive Critique." *Journal of Landscape Architecture* 13 (3): 12–15. https://doi.org/10.1080/18626033.2018.1589119.

Xiao, T., T. Mu, S. Shen, Y. Song, S. Yang, and J. He. 2022. "A Dynamic Physical-Distancing Model to Evaluate Spatial Measures for Prevention of Covid-19 Spread." *Physica A: Statistical Mechanics and Its Applications* 592:126734. https://doi.org/10.1016/j.physa.2021.126734.

Yang, D., E. Yurtsever, V. Renganathan, K. Redmill, and Ü. Özgüner. 2021. "A Vision-Based Social Distancing and Critical Density Detection System for COVID-19." *Sensors* 21 (13): 4608. https://doi.org/10.3390/s21134608 .

Yang, S., and H. Kim. 2021. "Urban Digital Twin Applications as a Virtual Platform of Smart City." *International Journal of Sustainable Building Technology and Urban Development* 12 (4): 363–379. https://doi.org/10.22712/susb.20210030.

Yin, C., J. Cao, B. Sun, and J. Liu. 2023. "Exploring Built Environment Correlates of Walking for Different Purposes: Evidence for Substitution." *Journal of Transport Geography* 106:103505. https://doi.org/10.1016/j.jtrangeo.2022.103505.

Zhang, J., X. Zhang, Y. Yang, and B. Zhou. 2020. "Study on the Influence of One-Way Street Optimization Design on Traffic Operation System." *Measurement and Control* 53 (7–8): 1107–1115. https://doi.org/10.1177/0020294020932366.

Zhang, Y., S. Wu, Z. Zhao, X. Yang, and Z. Fang. 2023. "An Urban Crowd Flow Model Integrating Geographic Characteristics." *Scientific Reports* 13 (1): 1695. https://doi.org/10.1038/s41598-023-29000-5.